

ECON3200 Tutorial 10 Recap and Solutions

The Time-Inconsistency Problem of Monetary Policy

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1. Recap Lectures 6, 7, 8, 9 and 10

1.1. Lecture 6: Money in the Short Run

Lecture 6 explains why money is non-neutral in the short run. The key reason is price and wage rigidities. Because prices and wages do not adjust immediately, monetary policy can affect real variables such as output and employment.

A contractionary monetary policy shock usually leads to

$$\hat{i}_t \uparrow \Rightarrow \hat{y}_t \downarrow \Rightarrow \hat{\pi}_t \downarrow \text{ slowly } \Rightarrow \hat{m}_t \downarrow.$$

The negative relationship between money supply and nominal interest rates is called the liquidity effect. Variables with hats are percentage deviations from steady state. For example, $\hat{u}_t < 0$ means u_t is below its steady-state value.

1.2. Lecture 7: Basic New Keynesian Model and Monetary Policy Shock

The basic New Keynesian model can be summarized by the NK IS curve, the NK Phillips curve, and the monetary policy rule:

$$\hat{x}_t = E_t[\hat{x}_{t+1}] - \frac{1}{\sigma} (\hat{i}_t - E_t[\hat{\pi}_{t+1}]) + u_t, \quad (1.NK^*)$$

$$\hat{\pi}_t = \beta E_t[\hat{\pi}_{t+1}] + \tilde{\kappa} \hat{x}_t, \quad (2.NK^*)$$

$$\hat{i}_t = \delta_\pi \hat{\pi}_t + \delta_x \hat{x}_t + v_t. \quad (3.NK^*)$$

The output gap is

$$\hat{x}_t = \hat{y}_t - \hat{y}_t^f.$$

Here $\hat{y}_t$ is actual output and $\hat{y}_t^f$ is flexible-price output. A monetary policy shock enters through

$$v_t = \rho_v v_{t-1} + \varepsilon_t.$$

A contractionary monetary policy shock means $\varepsilon_t > 0$, so

$$\varepsilon_t \uparrow \Rightarrow v_t \uparrow \Rightarrow \hat{i}_t \uparrow.$$

The real interest rate is

$$\hat{r}_t = \hat{i}_t - E_t[\hat{\pi}_{t+1}].$$

Since prices are sticky, expected inflation does not rise one-for-one with the nominal interest rate. Hence,

$$\hat{i}_t \uparrow \Rightarrow \hat{r}_t \uparrow \Rightarrow \hat{x}_t \downarrow \Rightarrow \hat{\pi}_t \downarrow.$$

1.3. Lecture 8: Technology Shock, Spending Shock and Cost-Push Shock

Lecture 8 studies three main shocks in the NK economy.

Technology shock. Productivity follows

$$a_t = \rho_a a_{t-1} + \omega_t, \quad \hat{z}_t = a_t.$$

Flexible-price output is

$$\hat{y}_t^f = \left(\frac{1 + \phi}{\sigma + \phi} \right) \hat{z}_t.$$

A positive technology shock raises both actual output and flexible-price output, but flexible-price output rises more. Hence,

$$\hat{x}_t = \hat{y}_t - \hat{y}_t^f \downarrow,$$

so inflation falls through the Phillips curve.

Spending / demand shock. A spending shock g_t follows

$$g_t = \rho_g g_{t-1} + \omega_t^g.$$

Flexible-price output becomes

$$\hat{y}_t^f = \left(\frac{1 + \phi}{\sigma + \phi} \right) \hat{z}_t + \left(\frac{1}{\sigma + \phi} \right) g_t.$$

The IS curve becomes

$$\hat{x}_t = E_t[\hat{x}_{t+1}] - \frac{1}{\sigma} \left(\hat{i}_t - E_t[\hat{\pi}_{t+1}] \right) + u_t + u_t^g, \quad (1.NK^{**})$$

where

$$u_t = E_t[\hat{y}_{t+1}^f] - \hat{y}_t^f, \quad u_t^g = \frac{1}{\sigma} (g_t - E_t[g_{t+1}]).$$

A positive demand shock usually raises the output gap and inflation:

$$g_t \uparrow \Rightarrow u_t^g \uparrow \Rightarrow \hat{x}_t \uparrow \Rightarrow \hat{\pi}_t \uparrow.$$

Cost-push shock. With a cost-push shock,

$$\hat{\pi}_t = \beta E_t[\hat{\pi}_{t+1}] + \tilde{\kappa} \hat{x}_t + \varepsilon_t^w. \quad (2.NK^{***})$$

A positive cost-push shock directly raises inflation:

$$\varepsilon_t^w \uparrow \Rightarrow \hat{\pi}_t \uparrow.$$

The Central Bank may then raise $\hat{i}_t$, reducing $\hat{x}_t$. This creates a trade-off between inflation stabilization and output stabilization.

1.4. Lecture 9: Open Economies, Policy Rules and Trade-off

In a small open economy, CPI inflation and the output gap are affected by the terms of trade:

$$\hat{s}_t = \hat{p}_t^f - \hat{p}_t^h, \quad \Delta \hat{s}_t = \hat{s}_t - \hat{s}_{t-1}.$$

CPI inflation is

$$\hat{\pi}_t^c = \hat{\pi}_t^h + \gamma \Delta \hat{s}_t.$$

Here $\gamma \in (0, 1)$ is the weight of foreign goods in domestic consumption. If $\Delta \hat{s}_t < 0$, foreign goods become relatively cheaper, so CPI inflation falls.

Lecture 9 also studies monetary policy rules. With

$$\hat{i}_t = \delta \hat{\pi}_t + v_t, \quad (3B.NK^*)$$

the Taylor Principle requires

$$\delta > 1.$$

If the Central Bank does not respond strongly enough to inflation, expectations can become self-fulfilling.

With cost-push shocks, inflation can be written as

$$\hat{\pi}_t = \tilde{\kappa} \sum_{k=0}^{\infty} \beta^k E_t[\hat{x}_{t+k}] + \varepsilon_t^w. \quad (2\text{TO.NK}^*)$$

This shows the trade-off:

stabilize inflation $\Rightarrow$ negative output gap,

but

stabilize output $\Rightarrow$ higher inflation.

1.5. Lecture 10: Time-Inconsistency Problem of Monetary Policy

Lecture 10 connects the long-run and short-run views of inflation. In the long run, inflation creates welfare costs. In the short run, surprise inflation can increase output and employment. This creates a time-inconsistency problem.

The Central Bank may promise low inflation, but after the public has formed expectations, it has an incentive to create surprise inflation to raise output. If the public understands this incentive, expected inflation rises. Then there is no output gain, but inflation is higher. This is the inflation bias.

The Barro and Gordon model uses the Lucas-type supply function:

$$y = y^n + a(\pi - \pi^e) + e.$$

Here y is actual output, y^n is natural output, π is actual inflation, π^e is expected inflation, $a > 0$ measures the effect of inflation surprises on output, and e is a supply shock.

The Central Bank's gain/objective function is

$$U = \lambda(y - y^n) - \frac{1}{2}\pi^2.$$

The Central Bank likes output above its natural level, but dislikes inflation.

Actual inflation is determined by

$$\pi = \zeta + v,$$

where ζ is the money growth instrument and v is a velocity shock.

Under rational expectations,

$$\pi^e = E(\pi).$$

Under discretion, the Central Bank cannot commit, and the policy creates an inflation bias. Under commitment, the Central Bank can commit to low inflation, so expected inflation is lower and expected utility is higher.

Reputation can help reduce the inflation bias. The Central Bank can sustain a lower inflation rate $\bar{\pi}$ if the cost of losing reputation is at least as large as the gain from cheating:

$$C(\bar{\pi}) \geq G(\bar{\pi}).$$

In Lecture 10,

$$G(\bar{\pi}) = \frac{1}{2}(\lambda a - \bar{\pi})^2,$$

and

$$C(\bar{\pi}) = \frac{1}{2}\beta \left((\lambda a)^2 - \bar{\pi}^2 \right).$$

The minimum sustainable inflation rate is found by solving

$$G(\pi_{\min}) = C(\pi_{\min}).$$

2. Tutorial 10 Full Questions, In-Class Guides, and Solutions

2.1. Exercise 10.1

Question

Consider an economy à la Barro and Gordon (1983), described by

$$y = y^n + a(\pi - \pi^e) + e,$$

$$U = \lambda(y - y^n) - \frac{1}{2}\pi^2,$$

$$\pi = \zeta + v,$$

$$\pi^e = E(\pi),$$

with

$$E(v) = 0, \quad \text{Var}(v) = \sigma_v^2,$$

and

$$E(e) = 0, \quad \text{Var}(e) = \sigma_e^2.$$

Let

$$\lambda = 0.5, \quad a = 0.1, \quad y^n = 1, \quad \sigma_v^2 = 0.0001.$$

Under discretion, we find

$$\zeta = \lambda a.$$

Under commitment, we have

$$\zeta = 0.$$

Compute π^e , $E(y)$, and $E(U)$ under discretion and commitment.

In-class solution guide

Given:

$$\lambda = 0.5, \quad a = 0.1, \quad y^n = 1, \quad \sigma_v^2 = 0.0001.$$

$$\lambda a = (0.5)(0.1) = \underline{\hspace{2cm}}.$$

Part A: Discretion

Under discretion:

$$\zeta = \lambda a.$$

Actual inflation is

$$\pi = \zeta + v = \lambda a + v.$$

Expected inflation:

$$\pi^e = E(\pi) = E(\lambda a + v) = \lambda a + E(v) = \underline{\hspace{2cm}}.$$

Expected output:

$$y = y^n + a(\pi - \pi^e) + e.$$

Taking expectations:

$$E(y) = y^n + a(E(\pi) - \pi^e) + E(e).$$

Since $E(\pi) = \pi^e$ and $E(e) = 0$,

$$E(y) = \underline{\hspace{2cm}}.$$

Expected utility:

$$E(U) = E \left[\lambda(y - y^n) - \frac{1}{2} \pi^2 \right].$$

Since $E(y - y^n) = 0$,

$$E(U) = -\frac{1}{2} E(\pi^2).$$

Under discretion:

$$\pi = \lambda a + v.$$

So:

$$E(\pi^2) = (\lambda a)^2 + \sigma_v^2.$$

Therefore:

$$E(U) = -\frac{1}{2} [(\lambda a)^2 + \sigma_v^2].$$

Substitute:

$$E(U) = -\frac{1}{2} [(\underline{\hspace{2cm}})^2 + 0.0001] = \underline{\hspace{2cm}}.$$

Part B: Commitment

Under commitment:

$$\zeta = 0.$$

Actual inflation is

$$\pi = v.$$

Expected inflation:

$$\pi^e = E(v) = \underline{\hspace{2cm}}.$$

Expected output:

$$E(y) = y^n + a(E(\pi) - \pi^e) + E(e) = \underline{\hspace{2cm}}.$$

Expected utility:

$$E(U) = -\frac{1}{2}E(v^2) = -\frac{1}{2}\sigma_v^2 = \underline{\hspace{2cm}}.$$

Final answers:

Discretion:

$$\pi^e = \underline{\hspace{2cm}}, \quad E(y) = \underline{\hspace{2cm}}, \quad E(U) = \underline{\hspace{2cm}}.$$

Commitment:

$$\pi^e = \underline{\hspace{2cm}}, \quad E(y) = \underline{\hspace{2cm}}, \quad E(U) = \underline{\hspace{2cm}}.$$

Intuition: Under discretion, expected inflation is positive because the public anticipates the Central Bank's incentive to create surprise inflation. But because expectations are rational, there is no average output gain. The result is inflation bias.

Solution

Under discretion,

$$\zeta = \lambda a = (0.5)(0.1) = 0.05.$$

Actual inflation is

$$\pi = \zeta + v = 0.05 + v.$$

Expected inflation is

$$\pi^e = E(\pi) = E(0.05 + v) = 0.05.$$

Expected output is

$$E(y) = E[y^n + a(\pi - \pi^e) + e] = y^n + a(E(\pi) - \pi^e) + E(e).$$

Since $E(\pi) = \pi^e$ and $E(e) = 0$,

$$E(y) = y^n = 1.$$

Expected utility is

$$E(U) = E\left[\lambda(y - y^n) - \frac{1}{2}\pi^2\right].$$

Since $E(y - y^n) = 0$,

$$E(U) = -\frac{1}{2}E(\pi^2).$$

Under discretion,

$$E(\pi^2) = (\lambda a)^2 + \sigma_v^2.$$

Therefore,

$$E(U) = -\frac{1}{2}[(0.05)^2 + 0.0001] = -\frac{1}{2}(0.0025 + 0.0001) = -0.0013.$$

So under discretion:

$\pi^e = 0.05,$	$E(y) = 1,$	$E(U) = -0.0013.$
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Under commitment,

$$\zeta = 0.$$

So

$$\pi = v.$$

Expected inflation is

$$\pi^e = E(v) = 0.$$

Expected output is

$$E(y) = y^n = 1.$$

Expected utility is

$$E(U) = -\frac{1}{2}\sigma_v^2 = -\frac{1}{2}(0.0001) = -0.00005.$$

So under commitment:

$\pi^e = 0,$	$E(y) = 1,$	$E(U) = -0.00005.$
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Explanation

Under discretion, the Central Bank creates positive expected inflation:

$$\pi^e = \lambda a = 0.05.$$

But because the public expects this, there is no average output gain:

$$E(y) = y^n.$$

Therefore, discretion only creates inflation bias. Under commitment, expected inflation is zero and there is still no output gain, but expected utility is higher because inflation is lower.

2.2. Exercise 10.2

Question

In a Barro and Gordon (1983) model, the gains and costs from deviating from $\bar{\pi}$ are given by

$$G(\bar{\pi}) = \frac{1}{2}(\lambda a - \bar{\pi})^2,$$

$$C(\bar{\pi}) = \frac{1}{2}\beta((\lambda a)^2 - \bar{\pi}^2).$$

Let

$$\lambda = 0.4, \quad a = 1, \quad \beta = 0.9.$$

Compute the gains and costs from deviating for

$$(i) \quad \bar{\pi} = 0.02,$$

and

$$(ii) \quad \bar{\pi} = 0.025.$$

Can those inflation rates be sustained by a reputation?

In-class solution guide

Given:

$$\lambda = 0.4, \quad a = 1, \quad \beta = 0.9.$$

So:

$$\lambda a = (0.4)(1) = \underline{\hspace{2cm}}.$$

Formulas:

$$G(\bar{\pi}) = \frac{1}{2}(\lambda a - \bar{\pi})^2 = \frac{1}{2}(0.4 - \bar{\pi})^2.$$

$$C(\bar{\pi}) = \frac{1}{2}\beta((\lambda a)^2 - \bar{\pi}^2) = \frac{1}{2}(0.9)(0.16 - \bar{\pi}^2).$$

Decision rule:

$$C(\bar{\pi}) \geq G(\bar{\pi}) \Rightarrow \text{can be sustained by reputation.}$$

$$G(\bar{\pi}) > C(\bar{\pi}) \Rightarrow \text{cannot be sustained by reputation.}$$

Case 1: $\bar{\pi} = 0.02$

Gain:

$$G(0.02) = \frac{1}{2}(0.4 - 0.02)^2 = \underline{\hspace{2cm}}.$$

Cost:

$$C(0.02) = \frac{1}{2}(0.9)(0.16 - (0.02)^2) = \underline{\hspace{2cm}}.$$

Compare:

$$G(0.02) - C(0.02) = \underline{\hspace{2cm}}.$$

Conclusion:

$$\bar{\pi} = 0.02 \quad \text{can / cannot be sustained?} \quad \underline{\hspace{2cm}}.$$

Case 2: $\bar{\pi} = 0.025$

Gain:

$$G(0.025) = \frac{1}{2}(0.4 - 0.025)^2 = \underline{\hspace{2cm}}.$$

Cost:

$$C(0.025) = \frac{1}{2}(0.9)(0.16 - (0.025)^2) = \underline{\hspace{2cm}}.$$

Compare:

$$G(0.025) - C(0.025) = \underline{\hspace{2cm}}.$$

Conclusion:

$$\bar{\pi} = 0.025 \quad \text{can / cannot be sustained?} \quad \underline{\hspace{2cm}}.$$

Intuition: A low inflation promise is credible only if the future cost of losing reputation is at least as large as the current gain from cheating.

Solution

Given

$$\lambda a = (0.4)(1) = 0.4.$$

The gain and cost functions are

$$G(\bar{\pi}) = \frac{1}{2}(0.4 - \bar{\pi})^2,$$

and

$$C(\bar{\pi}) = \frac{1}{2}(0.9)(0.16 - \bar{\pi}^2).$$

For $\bar{\pi} = 0.02$,

$$G(0.02) = \frac{1}{2}(0.4 - 0.02)^2 = \frac{1}{2}(0.38)^2 = 0.0722.$$

$$C(0.02) = \frac{1}{2}(0.9)(0.16 - (0.02)^2) = 0.0718.$$

Thus,

$$G(0.02) - C(0.02) = 0.00038 > 0.$$

The gain from deviating is larger than the cost of losing reputation. Therefore,

$\bar{\pi} = 0.02$ cannot be sustained by reputation.

For $\bar{\pi} = 0.025$,

$$G(0.025) = \frac{1}{2}(0.4 - 0.025)^2 = \frac{1}{2}(0.375)^2 = 0.0703.$$

$$C(0.025) = \frac{1}{2}(0.9)(0.16 - (0.025)^2) = 0.0717.$$

Thus,

$$G(0.025) - C(0.025) = -0.00141 < 0.$$

The cost of losing reputation is larger than the gain from deviating. Therefore,

$\bar{\pi} = 0.025$ can be sustained by reputation.

Explanation

The Central Bank deviates if the gain from cheating exceeds the cost of losing reputation:

$$G(\bar{\pi}) > C(\bar{\pi}).$$

If

$$C(\bar{\pi}) \geq G(\bar{\pi}),$$

then the Central Bank has no incentive to cheat, so the inflation rate can be sustained by reputation.

2.3. Exercise 10.3

Question

In a Barro and Gordon (1983) model, the gains and costs from deviating from $\bar{\pi}$ are given by

$$G(\bar{\pi}) = \frac{1}{2}(\lambda a - \bar{\pi})^2,$$

$$C(\bar{\pi}) = \frac{1}{2}\beta \left((\lambda a)^2 - \bar{\pi}^2 \right).$$

Let

$$\lambda = 0.5, \quad a = 1.$$

Compute the minimum inflation $\pi_{\min}$ that can be sustained by a reputation for

$$(i) \quad \beta = 0.6,$$

and

$$(ii) \quad \beta = 0.8.$$

In-class solution guide

Given:

$$\lambda = 0.5, \quad a = 1.$$

So:

$$\lambda a = (0.5)(1) = \underline{\hspace{2cm}}.$$

Minimum sustainable inflation condition:

$$G(\pi_{\min}) = C(\pi_{\min}).$$

Use:

$$G(\pi_{\min}) = \frac{1}{2}(0.5 - \pi_{\min})^2,$$

and

$$C(\pi_{\min}) = \frac{1}{2}\beta(0.5^2 - \pi_{\min}^2).$$

Set them equal:

$$\frac{1}{2}(0.5 - \pi_{\min})^2 = \frac{1}{2}\beta(0.25 - \pi_{\min}^2).$$

Cancel $\frac{1}{2}$:

$$(0.5 - \pi_{\min})^2 = \beta(0.25 - \pi_{\min}^2).$$

This becomes:

$$(1 + \beta)\pi_{\min}^2 - \pi_{\min} + 0.25(1 - \beta) = 0.$$

Case 1: $\beta = 0.6$

Substitute:

$$(1 + 0.6)\pi_{\min}^2 - \pi_{\min} + 0.25(1 - 0.6) = 0.$$

$$1.6\pi_{\min}^2 - \pi_{\min} + 0.1 = 0.$$

Solve the quadratic:

$$\pi_{\min} = \underline{\hspace{2cm}}.$$

Case 2: $\beta = 0.8$

Substitute:

$$(1 + 0.8)\pi_{\min}^2 - \pi_{\min} + 0.25(1 - 0.8) = 0.$$

$$1.8\pi_{\min}^2 - \pi_{\min} + 0.05 = 0.$$

Solve the quadratic:

$$\pi_{\min} = \underline{\hspace{2cm}}.$$

Final answers:

$$\beta = 0.6 : \quad \pi_{\min} = \underline{\hspace{2cm}}.$$

$$\beta = 0.8 : \quad \pi_{\min} = \underline{\hspace{2cm}}.$$

Intuition: A higher β means the Central Bank cares more about the future cost of losing reputation. Therefore, lower inflation can be sustained when β is higher.

Solution

The minimum inflation rate that can be sustained by reputation is found at

$$G(\pi_{\min}) = C(\pi_{\min}).$$

Given

$$\lambda = 0.5, \quad a = 1,$$

we have

$$\lambda a = 0.5.$$

Therefore,

$$\frac{1}{2}(0.5 - \pi_{\min})^2 = \frac{1}{2}\beta(0.5^2 - \pi_{\min}^2).$$

Cancel $\frac{1}{2}$:

$$(0.5 - \pi_{\min})^2 = \beta(0.25 - \pi_{\min}^2).$$

Expand:

$$0.25 - \pi_{\min} + \pi_{\min}^2 = 0.25\beta - \beta\pi_{\min}^2.$$

Move all terms to the left:

$$(1 + \beta)\pi_{\min}^2 - \pi_{\min} + 0.25(1 - \beta) = 0.$$

For $\beta = 0.6$, the equation becomes

$$1.6\pi_{\min}^2 - \pi_{\min} + 0.1 = 0.$$

Solving this quadratic gives two roots:

$$\pi_{\min} = 0.125 \quad \text{and} \quad \pi = 0.5.$$

The minimum sustainable inflation is the smaller root:

$$\boxed{\pi_{\min} = 0.125.}$$

For $\beta = 0.8$, the equation becomes

$$1.8\pi_{\min}^2 - \pi_{\min} + 0.05 = 0.$$

Solving this quadratic gives two roots:

$$\pi_{\min} = 0.0555 \quad \text{and} \quad \pi = 0.5.$$

The minimum sustainable inflation is the smaller root:

$$\boxed{\pi_{\min} = 0.0555.}$$

Explanation

The minimum sustainable inflation rate is the point where the gain from deviating equals the cost of losing reputation:

$$G(\pi_{\min}) = C(\pi_{\min}).$$

If inflation is below this level, the gain from cheating is too large, so the Central Bank will deviate. If inflation is at or above this level, reputation can sustain it.

A higher β means the Central Bank values the future more. Therefore, the cost of losing reputation is

larger, and the minimum sustainable inflation is lower:

$$\beta \uparrow \Rightarrow \pi_{\min} \downarrow .$$

3. Main Takeaway

Lecture 10 explains the time-inconsistency problem of monetary policy. The Central Bank may promise low inflation, but once the public has formed expectations, it has an incentive to create surprise inflation to increase output. Under rational expectations, the public anticipates this incentive, so expected inflation rises. There is no average output gain, but inflation is higher. This is the inflation bias.

Under commitment:

$$\pi^e = 0, \quad E(y) = y^n,$$

and expected utility is higher because inflation is lower.

Under discretion:

$$\pi^e = \lambda a, \quad E(y) = y^n,$$

so there is no output gain, but there is inflation bias.

Reputation can reduce the inflation bias if

$$C(\bar{\pi}) \geq G(\bar{\pi}).$$

The minimum sustainable inflation rate is found from

$$G(\pi_{\min}) = C(\pi_{\min}).$$

A more patient Central Bank, with higher β , can sustain lower inflation:

$$\beta \uparrow \Rightarrow \pi_{\min} \downarrow .$$